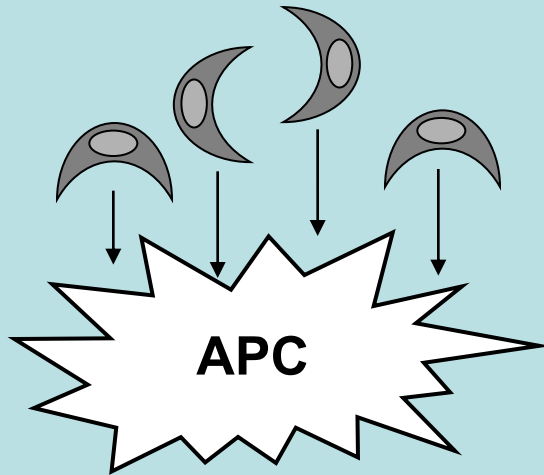


IL-12/IL-4 Independent Th Polarization by Dendritic Cells Exposed to Extracts from Th1/Th2 Pathogens

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**Intracellular
protozoa**

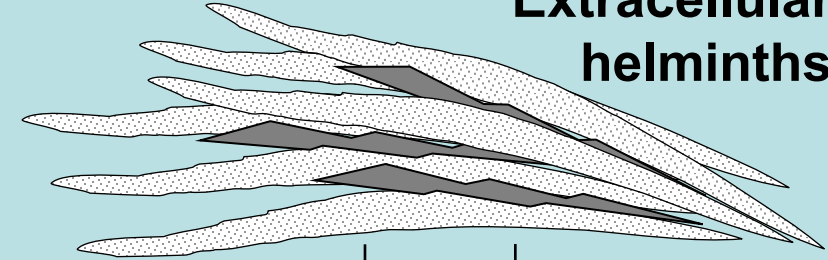


IL-12



**Th1
effector**

**Extracellular
helminths**

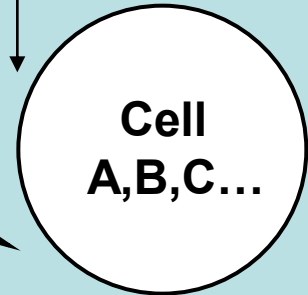


IL-4



**Th2
effector**

**Cell
A,B,C...**

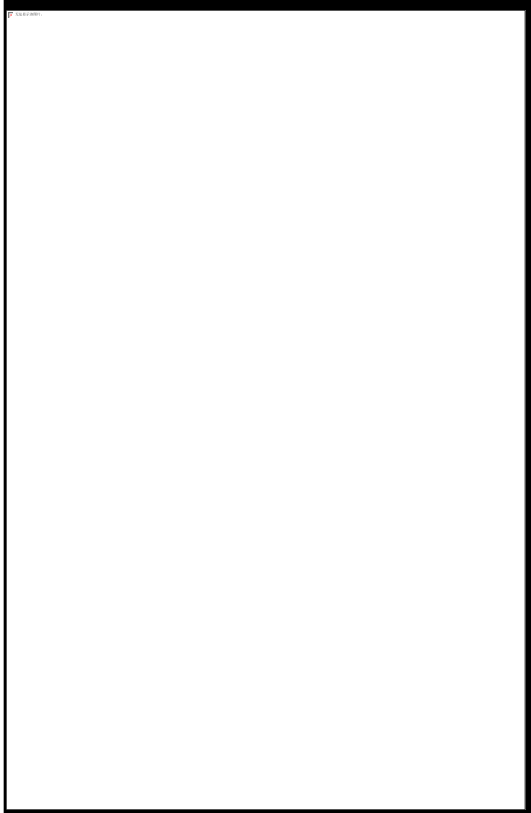


Questions:

What is the precise role of the polarizing cytokines in Th effector choice during parasitic infections?

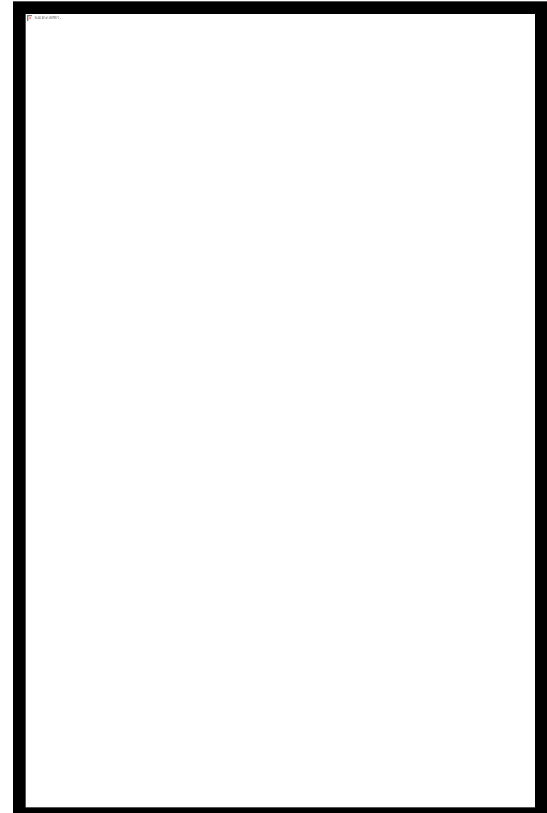
**Are IL-12/IL-4 required for the initial Th1/Th2
_____ decision step or are they merely stabilizing
effector populations that have already emerged?**

Toxoplasma gondii
tachyzoites



polarized
Th1 response

Schistosoma mansoni
ova



polarized
Th2 response

Conclusions from previous studies:

IL-4 is not required for in vivo development of Th2 cells in response to *S. mansoni* infection, but plays an important role in stabilizing their phenotype and negatively regulating the generation of Th1 cells.

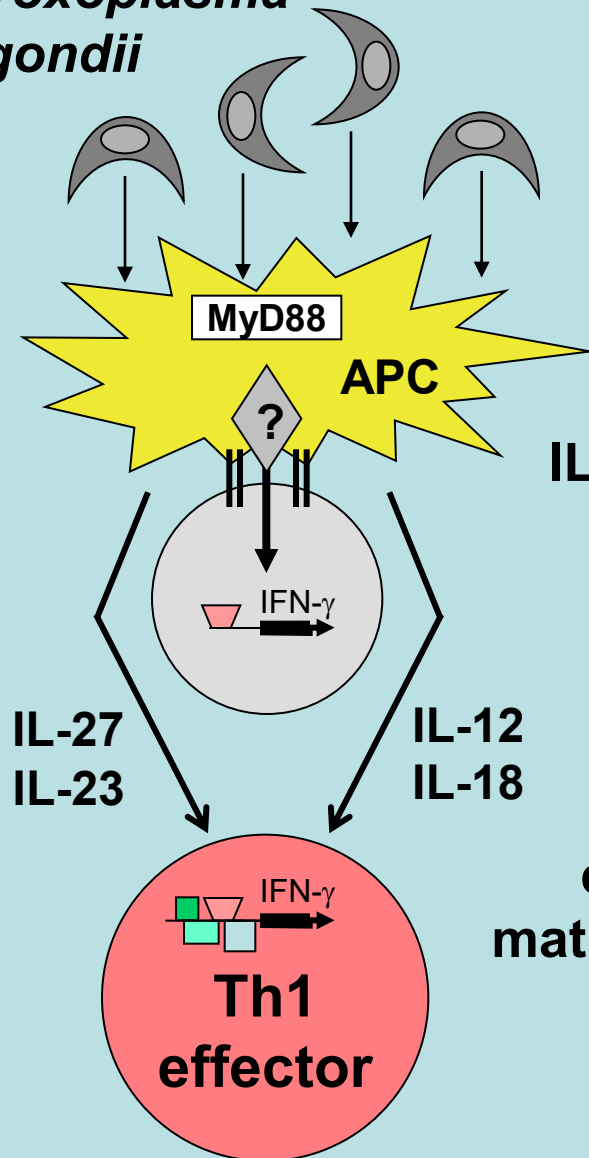
Jankovic et al. J. Immunol. 2000.

Similarly, *T. gondii* is able to stimulate Th1 development in the absence of IL-12. While no default to a Th2 pattern is observed in *T. gondii* Ag primed IL-12 KO mice, a complete default to a Th2 cytokine pattern is observed in MyD88 KO animals.

Jankovic et al. Immunity 2002.

Toxoplasma gondii

Schistosoma mansoni

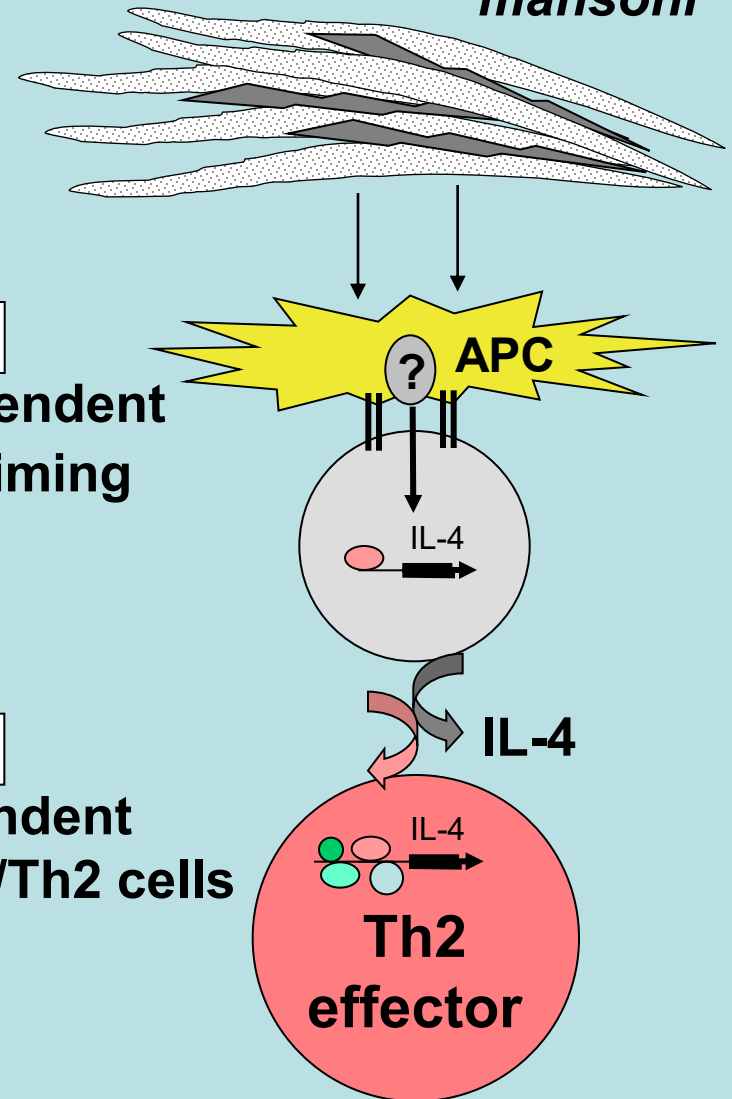


Step 1:

IL-12/IL-4 independent
CD4⁺ T cell priming

Step 2:

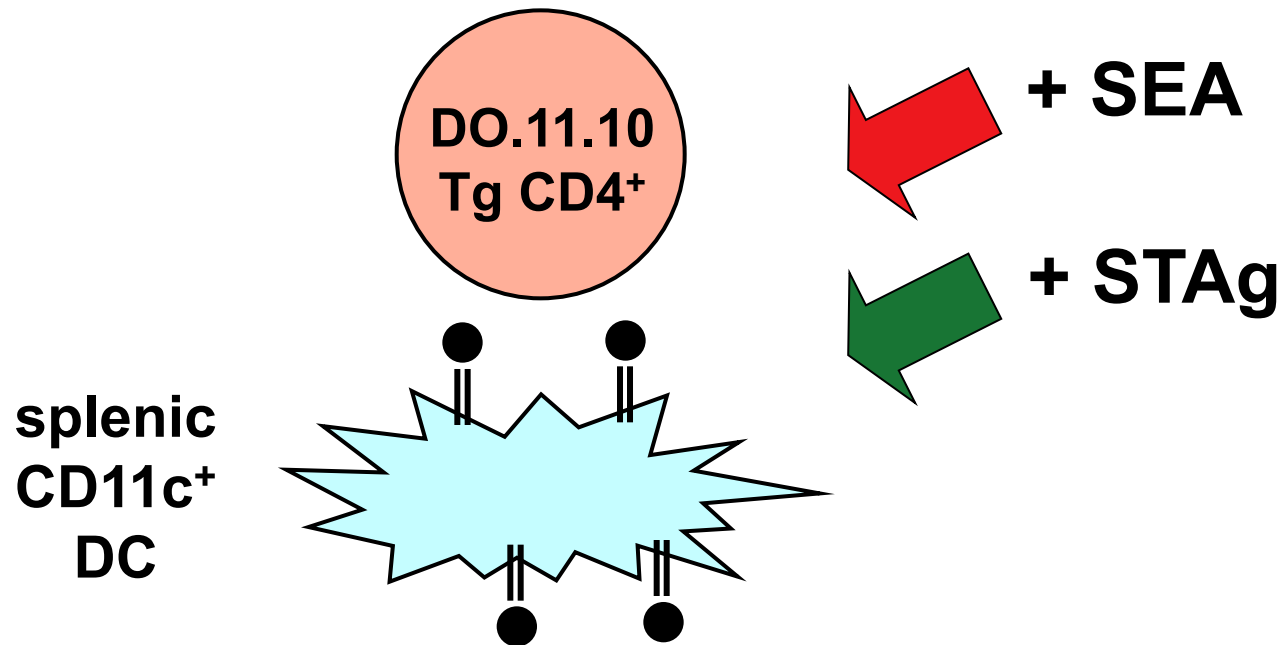
cytokine dependent
maturation of Th1/Th2 cells



Question:

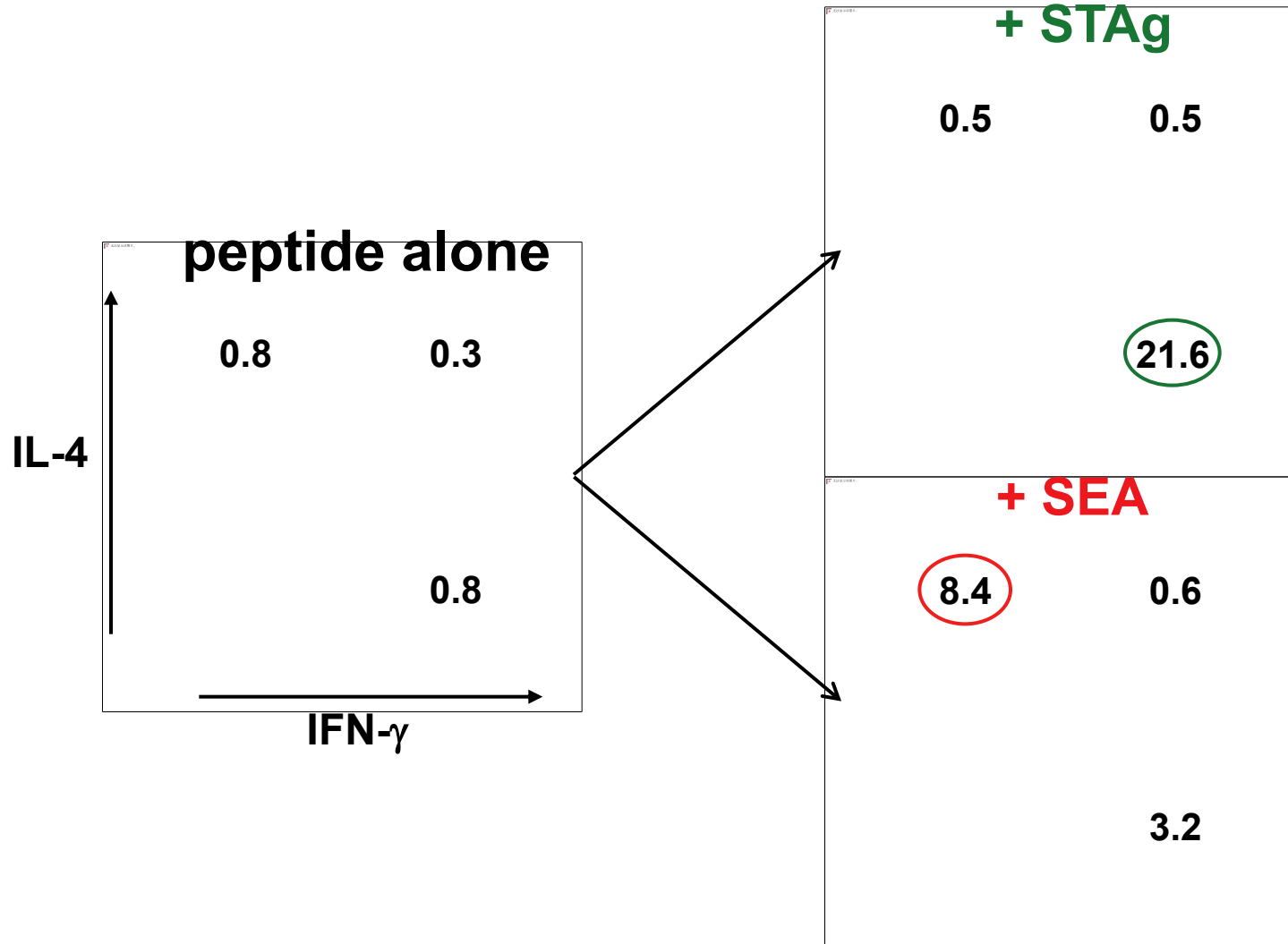
Are dendritic cells conditioned by exposure to parasite extracts able to promote IL-12/IL-4 independent Th1/Th2 commitment?

In Vitro Experimental Model

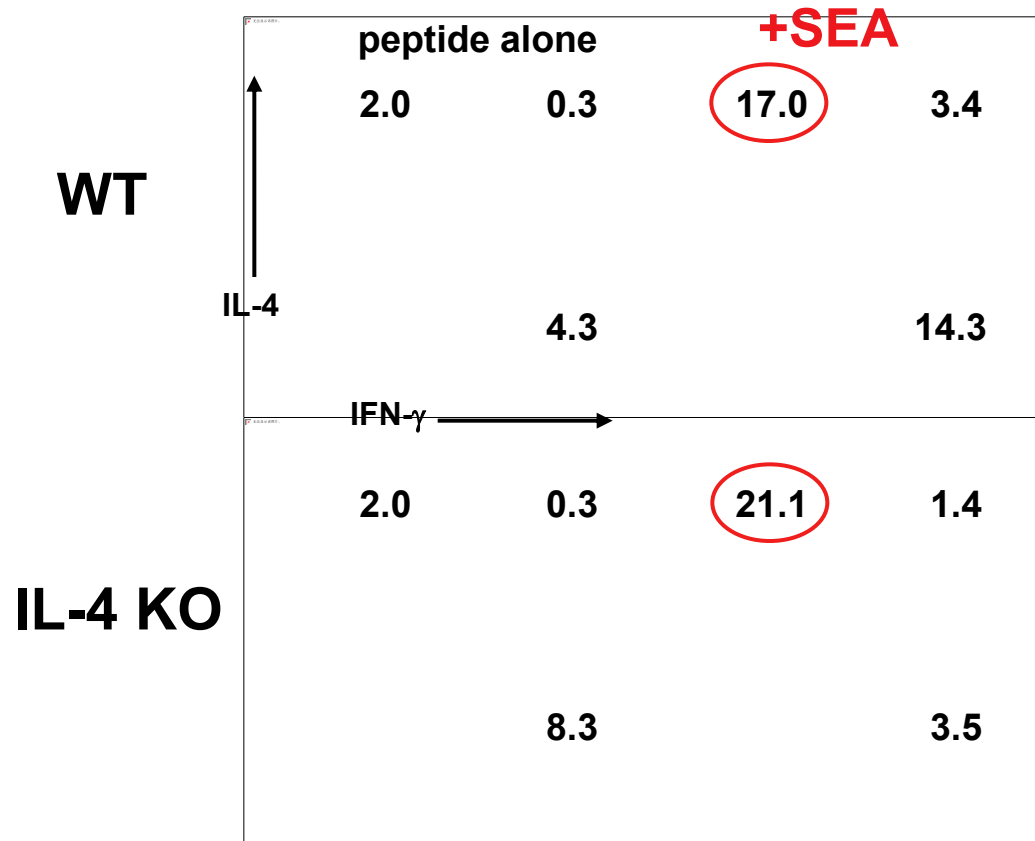


After 7 d-culture intracellular cytokine staining is performed, and phenotype of CD4⁺ cells is determined.

Th polarization induced by dendritic cells exposed to extracts from Th1/Th2 pathogens



SEA-primed DC induce Th2 polarization



STAg-primed DC induce Th1 polarization independently of IL-12 secretion

		peptide alone		+STAg		+SEA	
WT	IL-4 ↑	1.4	0.3	0.6	0.7	13.6	3.1
			8.6		38.3		14.6
12 KO	IFN-γ →	1.5	0.2	1.5	0.8	16.2	3.3
			4.5		21.1		10.7
irradiated 12 KO		9.9	0.5	8.2	0.5	13.3	1.3
			1.0		1.2		2.6

STAg-induced Th1 priming dominates over SEA-induced Th2 priming even in the absence of IL-12

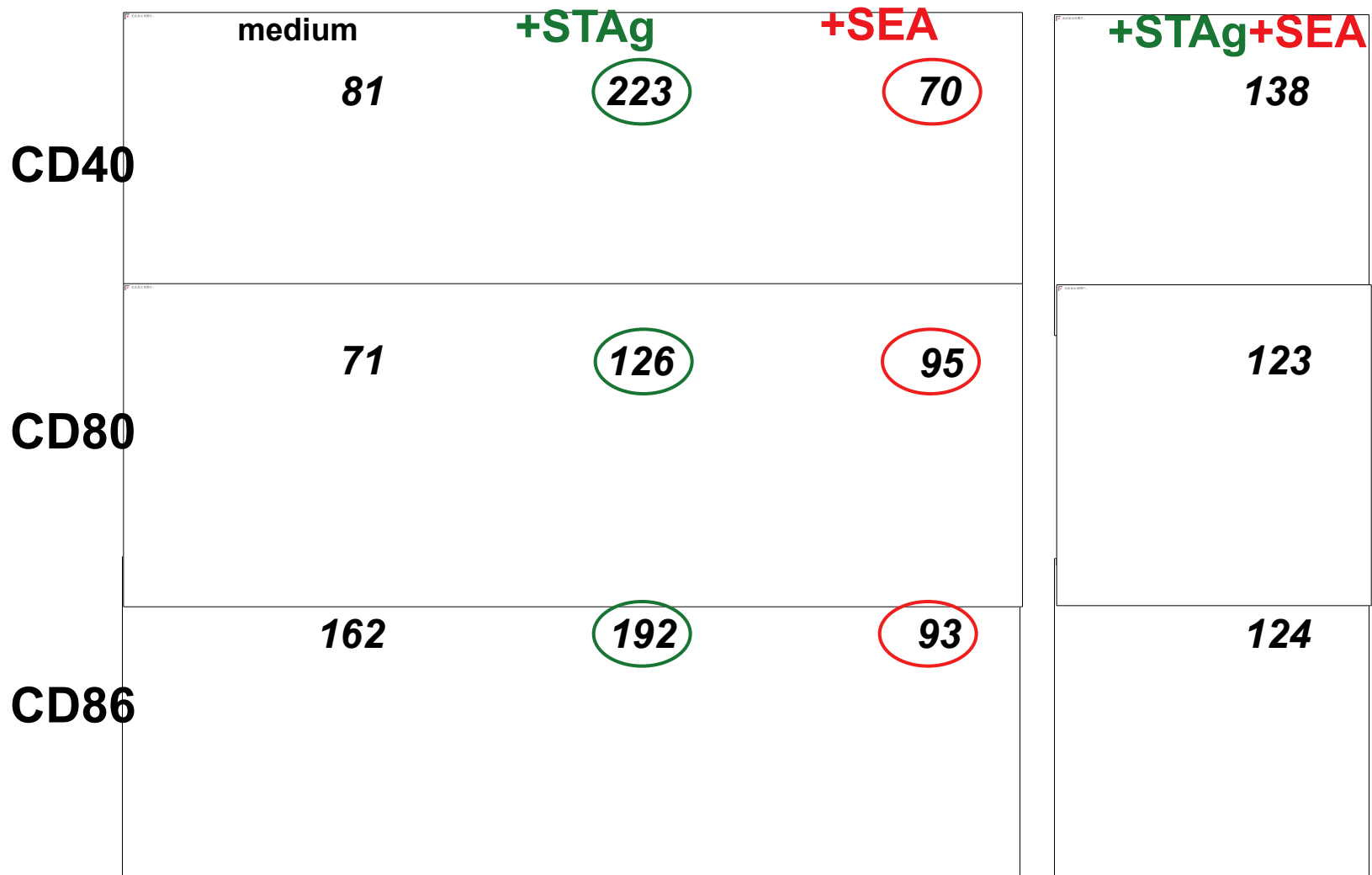
		peptide alone		+STAg		+SEA			
WT	IL-4 ↑	1.4	0.3	0.6	0.7	13.6	3.1	+STAg+SEA	
								1.2	1.2
12 KO	IFN-γ →		8.6		38.3		14.4		
		1.5	0.2	1.5	0.8	16.3	3.3	3.2	3.4
			4.5		21.1		10.7		
								32.5	

Question:

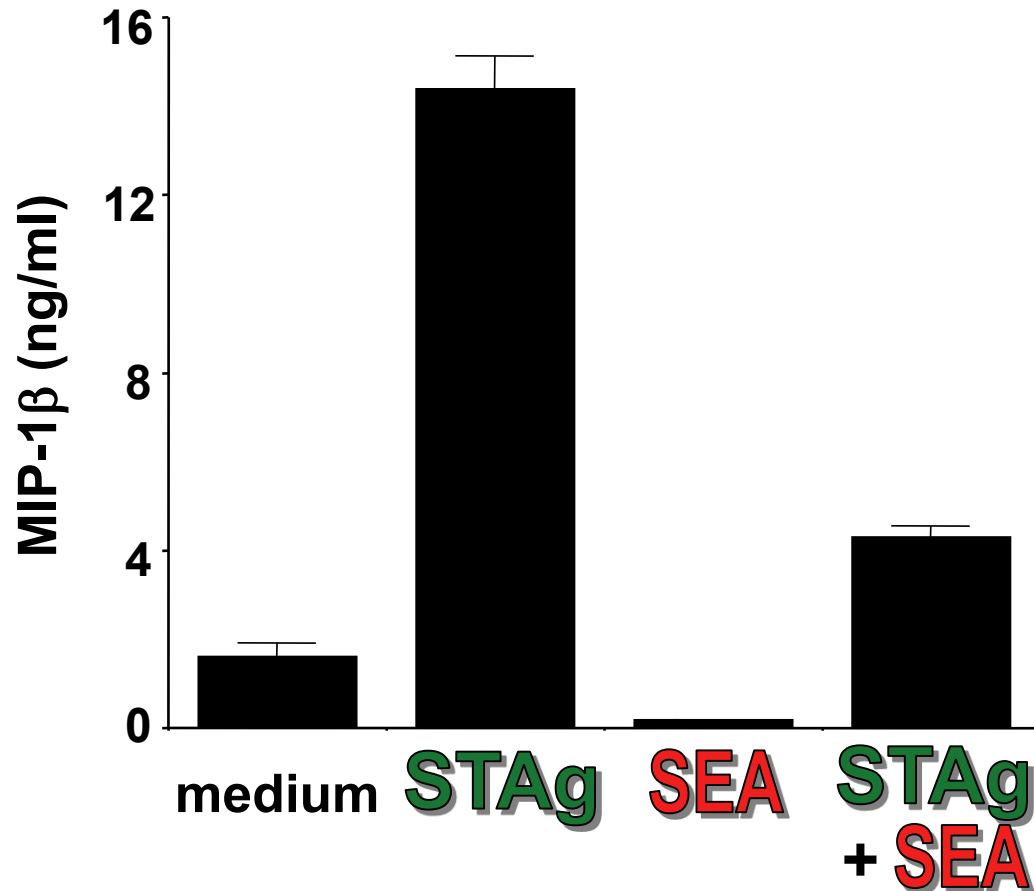
What is the effect of STAg and SEA on DC function:

- a) expression of costimulatory molecules?**
- b) chemokine expression?**

SEA down-regulates STAg-induced CD40 and CD86 expression on DC



SEA down-regulates MIP-1 β secretion by DC



Conclusions :

- **DC exposed to parasite extracts promote Th1/Th2 commitment in an IL-12/IL-4 independent fashion.**
- **The opposing effects of STAg and SEA on Th phenotype correlate with distinct states of DC activation:**
 - **STAg exposure induces upregulation of costimulatory molecules and chemokine production in DC.**
 - **In contrast, exposure to SEA results in down-regulation of many of these functions.**

These findings suggest that Th2 development depends not on a default resulting from the absence of DC activation, but from an actual suppression of DC function.